

Logistic Regression

Solved Examples

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Derivation of Sigmoid Function

- To understand logistic regression, let's go over the odds of success.

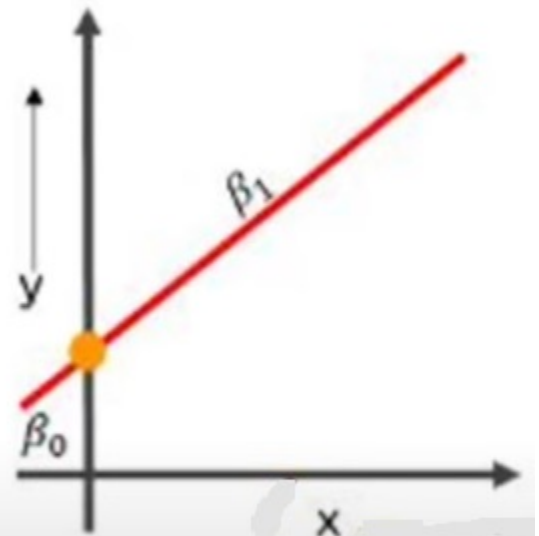
- $$\text{Odds}(\theta) = \frac{\text{Probability of an event happening}}{\text{Probability of an event not happening}}$$

- $$\text{Odds}(\theta) = \frac{p}{1-p}$$

- The values of odds range from **zero to ∞** and the values of probability lies between **zero and one**.

- Consider the equation of a straight line:

- $$y = \beta_0 + \beta_1 * x$$



- Now to predict the odds of success, we take log on odds formula:

$$\log \left(\frac{p(x)}{1-p(x)} \right) = \beta_0 + \beta_1 x$$

$$\ln \left(\frac{P(Y)}{1-P(Y)} \right) = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_K X_K$$

↑ **intercept**
↑ **model coefficients**
↑
↑
↑

- Exponentiating both the sides, we have:

$$e^{\ln \left(\frac{p(x)}{1-p(x)} \right)} = e^{\beta_0 + \beta_1 x}$$

$$e^{\ln(x)} = x$$

$$\left(\frac{p(x)}{1-p(x)} \right) = e^{\beta_0 + \beta_1 x}$$

- Let $Y = e^{\beta_0 + \beta_1 x}$

- Then $\frac{p(x)}{1 - p(x)} = Y$

- $p(x) = Y(1 - p(x))$

- $p(x) = (Y - Yp(x))$

- $p(x) + Y(p(x)) = Y$

- $p(x)(1 + Y) = Y$

- $p(x) = \frac{Y}{1+Y}$

$$p(x) = \frac{e^{\beta_0 + \beta_1 x}}{1 + e^{\beta_0 + \beta_1 x}}$$

Divide every term by: $e^{\beta_0 + \beta_1 x}$

The equation of the sigmoid function is:

$$p(x) = \frac{1}{1 + e^{-(\beta_0 + \beta_1 x)}}$$

Step-by-Step Flow

Model makes prediction

$$z = wx + b \quad , \quad a = \sigma(z)$$

Output: predicted probability (a)

Compute Log Loss

$$L = -\frac{1}{n} \sum [y \log(a) + (1 - y) \log(1 - a)]$$

Output: how wrong the model is

Compute Gradient of Loss

$$dw = (a - y)x \quad , \quad db = (a - y)$$

Output: direction + magnitude of correction

Apply Gradient Descent

$$w = w - \alpha \cdot dw$$

$$b = b - \alpha \cdot db$$

Output: updated (better) parameters

Input (x, y)



Prediction ($a = \sigma(wx+b)$)



Compute Log Loss



Compute Gradient (dw, db)



Update Parameters (w, b)



Repeat 

Iteration 1

- $x_1 = \text{Age (scaled)}$
- $x_2 = \text{Risk count (Diabetes + Hypertension + Smoker + Surgery)}$

Age	Risk Count	y
1	1	0
2	1	0
3	1	0
4	2	1
5	3	1

$$z = w_1x_1 + w_2x_2 + b$$

single output node, logistic sigmoid activation $\sigma(z) = \frac{1}{1 + e^{-z}}$

Initial Values

- $w_1 = 0$
- $w_2 = 0$
- $b = 0$
- Learning rate $\alpha = 0.1$

Since all weights = 0:

$$z = 0 \Rightarrow a = 0.5$$

Example (Sigmoid of 0)

- Formula: $\frac{1}{1+e^{-0}}$
- Input: 1 ÷ (1 + SHIFT ln (-) 0) =
- Result: 0.5

- Calculate Log Loss

$$L = -\frac{1}{n} \sum [y \log(a) + (1 - y) \log(1 - a)]$$

y value	Formula simplifies to
y = 0	$-\log(1 - a)$
y = 1	$-\log(a)$

- Compute Gradients (for all samples)

$$dw = \frac{1}{n} \sum (a - y)x, \quad db = \frac{1}{n} \sum (a - y)$$

Compute loss

For both cases:

- If $y = 0$:

$$-\log(1 - 0.5) = -\log(0.5) = 0.693$$

- If $y = 1$:

$$-\log(0.5) = 0.693$$

y	a	(a - y)
0	0.5	0.5
0	0.5	0.5
0	0.5	0.5
1	0.5	-0.5
1	0.5	-0.5

Average Loss

$$L_1 = (0.693 + 0.693 + 0.693 + 0.693 + 0.693) / 5$$

$$L_1 = 0.693$$

Compute gradients and Apply Gradient Decent

For w_1 (Age)

$$\begin{aligned}dw_1 &= \frac{1}{5}[(0.5)(1) + (0.5)(2) + (0.5)(3) + (-0.5)(4) + (-0.5)(5)] \\&= \frac{1}{5}(0.5 + 1 + 1.5 - 2 - 2.5) = \frac{-1.5}{5} = -0.3\end{aligned}$$

$$w = w - \alpha(a - y)x$$

$$b = b - \alpha(a - y)$$

For w_2 (Risk)

$$\begin{aligned}dw_2 &= \frac{1}{5}[(0.5)(1) + (0.5)(1) + (0.5)(1) + (-0.5)(2) + (-0.5)(3)] \\&= \frac{1}{5}(0.5 + 0.5 + 0.5 - 1 - 1.5) = \frac{-1}{5} = -0.2\end{aligned}$$

Update

$$w_1 = 0 - 0.1(-0.3) = 0.03$$

$$w_2 = 0 - 0.1(-0.2) = 0.02$$

$$b = 0 - 0.1(0.1) = -0.01$$

Bias

$$db = \frac{1}{5}(0.5 + 0.5 + 0.5 - 0.5 - 0.5) = 0.1$$

ITERATION 2

Now weights are:

- $w_1 = 0.03$
- $w_2 = 0.02$
- $b = -0.01$

Compute z , a , and error

$$z = w_1x_1 + w_2x_2 + b \quad , \quad \sigma(z) = \frac{1}{1 + e^{-z}}$$

(x_1)	(x_2)	(y)	(z) calculation	(z)	$\sigma(z) = \frac{1}{1 + e^{-z}}$	$(a - y)$
1	1	0	$0.03(1)+0.02(1)-0.01$	0.04	0.510	0.510
2	1	0	$0.06+0.02-0.01$	0.07	0.517	0.517
3	1	0	$0.09+0.02-0.01$	0.10	0.525	0.525
4	2	1	$0.12+0.04-0.01$	0.15	0.537	-0.463
5	3	1	$0.15+0.06-0.01$	0.20	0.550	-0.450

Row 1 ($y = 0, a = 0.510$)

$$\text{Loss} = -\log(1 - 0.510) = -\log(0.490) \approx 0.713$$

Row 2 ($y = 0, a = 0.517$)

$$-\log(0.483) \approx 0.728$$

Row 3 ($y = 0, a = 0.525$)

$$-\log(0.475) \approx 0.744$$

Row 4 ($y = 1, a = 0.537$)

$$-\log(0.537) \approx 0.622$$

Row 5 ($y = 1, a = 0.550$)

$$-\log(0.550) \approx 0.598$$

Log Loss Calculation

$$L = -\frac{1}{n} \sum [y \log(a) + (1 - y) \log(1 - a)]$$

y value	Formula simplifies to
$y = 0$	$-\log(1 - a)$
$y = 1$	$-\log(a)$

Average Loss

$$L = \frac{0.713 + 0.728 + 0.744 + 0.622 + 0.598}{5}$$

$$L = \frac{3.405}{5} \approx 0.681$$

Final Log Loss ≈ 0.681

Compute Gradients

For w_1

$$\begin{aligned}dw_1 &= \frac{1}{5} \sum (a - y)x_1 \\&= \frac{1}{5} [(0.510)(1) + (0.517)(2) + (0.525)(3) + (-0.463)(4) + (-0.450)(5)] \\&= \frac{1}{5} (0.510 + 1.034 + 1.575 - 1.852 - 2.250) \\&= \frac{-0.983}{5} = -0.197\end{aligned}$$

For w_2

$$\begin{aligned}dw_2 &= \frac{1}{5} \sum (a - y)x_2 \\&= \frac{1}{5} [(0.510)(1) + (0.517)(1) + (0.525)(1) + (-0.463)(2) + (-0.450)(3)] \\&= \frac{1}{5} (0.510 + 0.517 + 0.525 - 0.926 - 1.350) \\&= \frac{-0.724}{5} = -0.145\end{aligned}$$

Compute gradients and Apply Gradient Decent

For bias

$$\begin{aligned}db &= \frac{1}{5} (0.510 + 0.517 + 0.525 - 0.463 - 0.450) \\&= \frac{0.639}{5} = 0.128\end{aligned}$$

Update Parameters

$$w_1 = 0.03 - 0.1(-0.197) = 0.0497$$

$$w_2 = 0.02 - 0.1(-0.145) = 0.0345$$

$$b = -0.01 - 0.1(0.128) = -0.0228$$

Parameter	Value
(w_1)	0.0497
(w_2)	0.0345
(b)	-0.0228

A hospital wants to predict the likelihood of patients developing complications after a surgical procedure based on various factors such as age, pre-existing health conditions, and type of surgery. They have collected data on 500 patients who underwent the same surgical procedure in the past year. For each patient, they recorded their age, presence of diabetes, presence of hypertension, whether they smoke, and the type of surgery performed (coded as 1 for minor surgery and 2 for major surgery), as well as whether they experienced complications post-surgery (coded as 1 for yes and 0 for no).

Age	Diabetes	Hypertension	Smoker	Surgery Type	Complications
45	0	1	1	1	0
60	1	1	0	2	1
55	1	0	1	1	1
70	0	1	0	2	1
50	1	1	1	1	0

Smoker (b4)	1.0
Surgery Type (b5)	2.0

Step 3: New Patient Information

Feature	Value
Age	60
Diabetes	1
Hypertension	1
Smoker	1
Surgery Type	2

use the model to predict the probability of complications for a new patient who is 60 years old, has diabetes and hypertension, is a smoker, and is undergoing major surgery.

The logistic regression model is:

$$P(Y = 1) = 1 / 1 + e^{-(b_0 + b_1x_1 + b_2x_2 + b_3x_3 + b_4x_4 + b_5x_5)}$$

Step 2: Example Coefficients

Assume the trained logistic regression model produced these coefficients (illustrative example):

Variable	Coefficient
Intercept (b0)	-8
Age (b1)	0.05
Diabetes (b2)	1.5
Hypertension (b3)	1.2

Step 4: Compute Linear Combination (z)

$$z = -8 + (0.05 \times 60) + (1.5 \times 1) + (1.2 \times 1) + (1.0 \times 1) + (2 \times 2)$$

$$z = -8 + 3 + 1.5 + 1.2 + 1 + 4, z = 2.7$$

Step 5: Convert to Probability

$$P(Y = 1) = 1 / 1 + e^{-2.7}$$

$$e^{-2.7} = 0.067$$

$$P(Y = 1) = 1 / 1 + 0.067$$

$$P(Y = 1) = 0.937$$

1. Calculate the probability of pass for the student who studied 33 hours.

$$\bullet \quad p = \frac{1}{1+e^{-z}} \qquad s(x) = \frac{1}{1+e^{-x}}$$

$$\bullet \quad z = -64 + 2 * 33 = -64 + 66 = 2$$

$$\bullet \quad p = \frac{1}{1+e^{-2}} = 0.88$$

- That is, if student studies 33 hours, then is **88% chance** that the student will pass the exam

Hours Study	Pass (1) / Fail (0)
29	0
15	0
33	1
28	1
39	1

$$\log(odds) = z = -64 + 2 * hours$$

2. At least how many hours student should study that makes he will pass the course with the probability of more than 95%.

- $p = \frac{1}{1+e^{-z}} = 0.95$
- $0.95 * (1 + e^{-z}) = 1$
- $0.95 * e^{-z} = 1 - 0.95$
- $e^{-z} = \frac{0.05}{0.95} = 0.0526$
- $\ln(e^{-z}) = \ln(0.0526)$

$$\ln(e^x) = x$$

$$-z = \ln(0.0526) = -2.94$$

$$z = 2.94$$

- $z = 2.94$
- $\log(odds) = z = -64 + 2 * hours$
- $2.94 = -64 + 2 * hours$
- $2 * hours = 2.94 + 64$
- $2 * hours = 66.94$
- $hours = \frac{66.94}{2}$
- **$hours = 33.47 \text{ Hours}$**